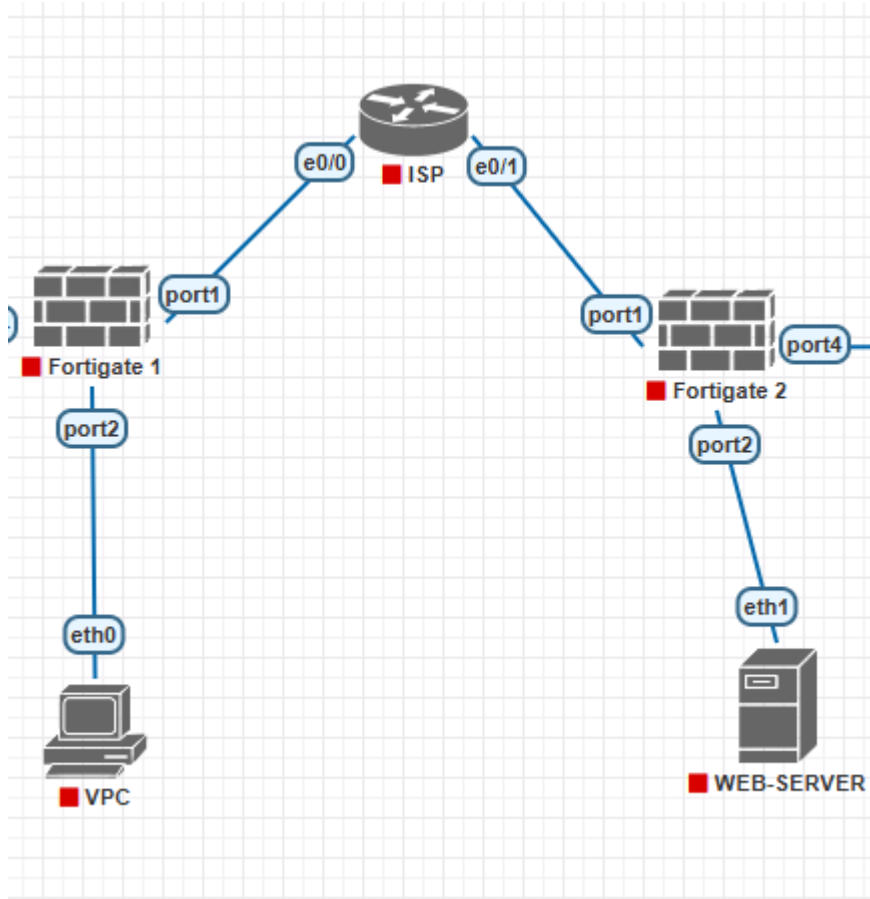


Topología 1

El proposito de esta topologia es poder establecer una conexión vpn entre 2 firewall fortigate.



Matrícula: 2025-1410

La ip base se basara en los últimos 4 dígitos de la matricula

Red base: 14.10.0.0/16

Subred	Dirección de Red	Máscara	Prefijo	Hosts Usables	Rango de Host Útil	Broadcast	Uso / Asignación
1	14.10.1.0	255.255.255.252	/30	2	14.10.1.1 – 14.10.1.2	14.10.1.3	ISP ↔ FortiGate 1 (.1 =

							FGT1, .2 = ISP)
2	14.10.2.0	255.255.255.252	/30	2	14.10.2.1 – 14.10.2.2	14.10.2.3	ISP ↔ FortiGate 2 (.1 = FGT2, .2 = ISP)
3	14.10.10.0	255.255.255.128	/25	126	14.10.10.1 – 14.10.10.126	14.10.10.127	LAN Usuarios / VLAN10 (.1 = Gateway FGT1)
4	14.10.20.0	255.255.255.240	/28	14	14.10.20.1 – 14.10.20.14	14.10.20.15	LAN Servidor Web (.1 = Gateway FGT2, .2 = Server)

Fortigate 1

FortiGate - FortiGate-VM64-KVM x

Not secure 192.168.189.153/ng/interface

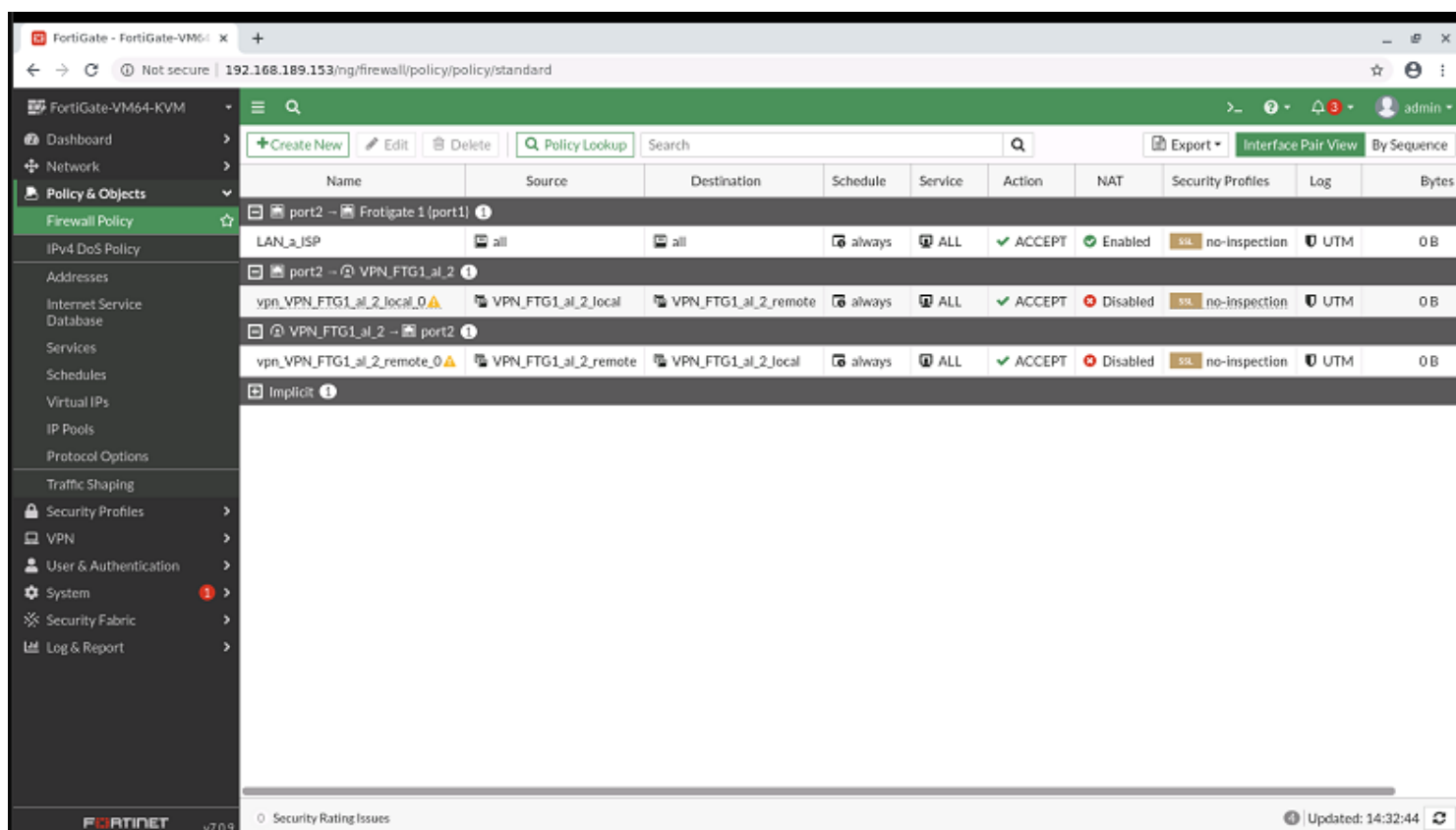
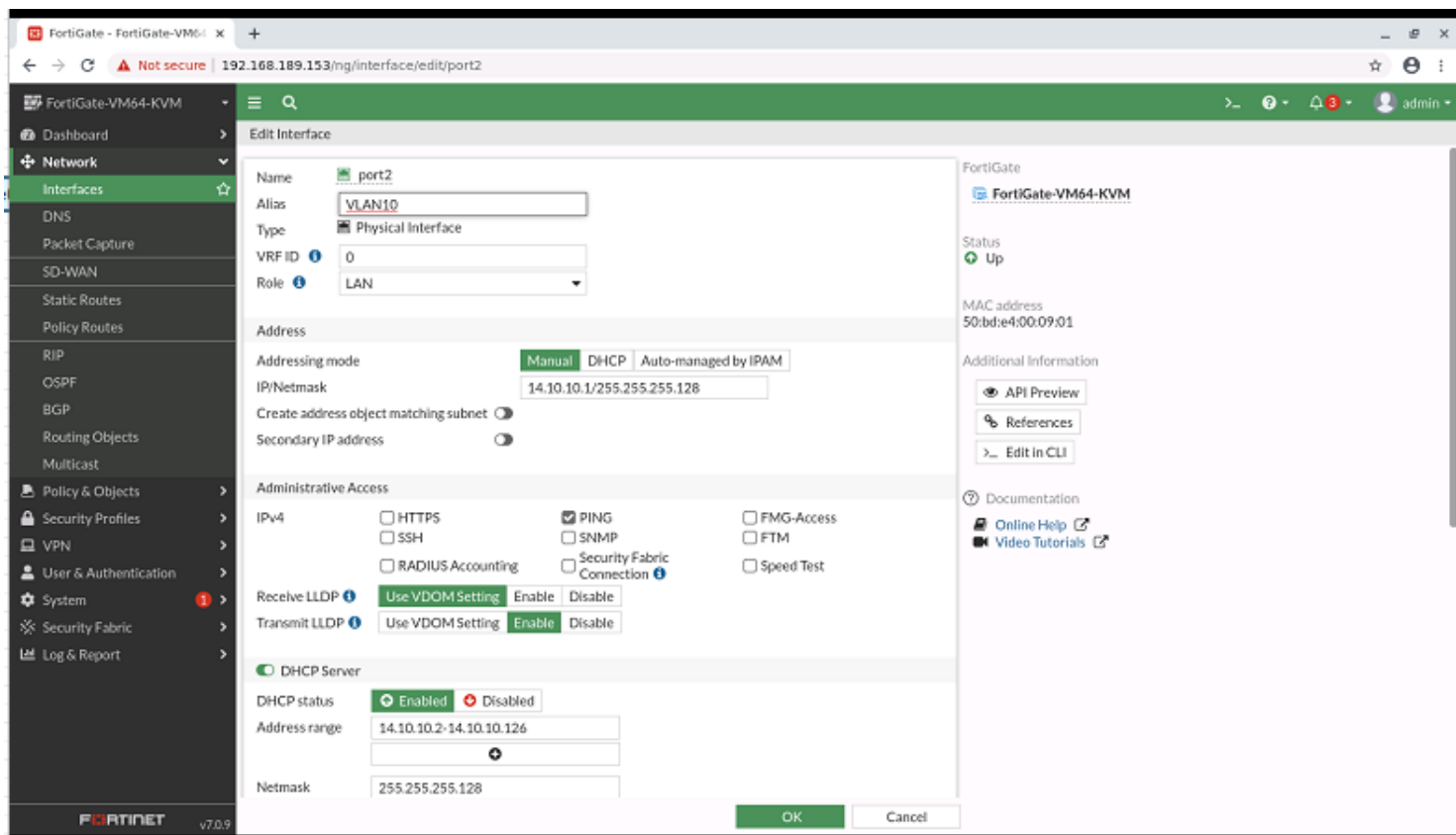
admin

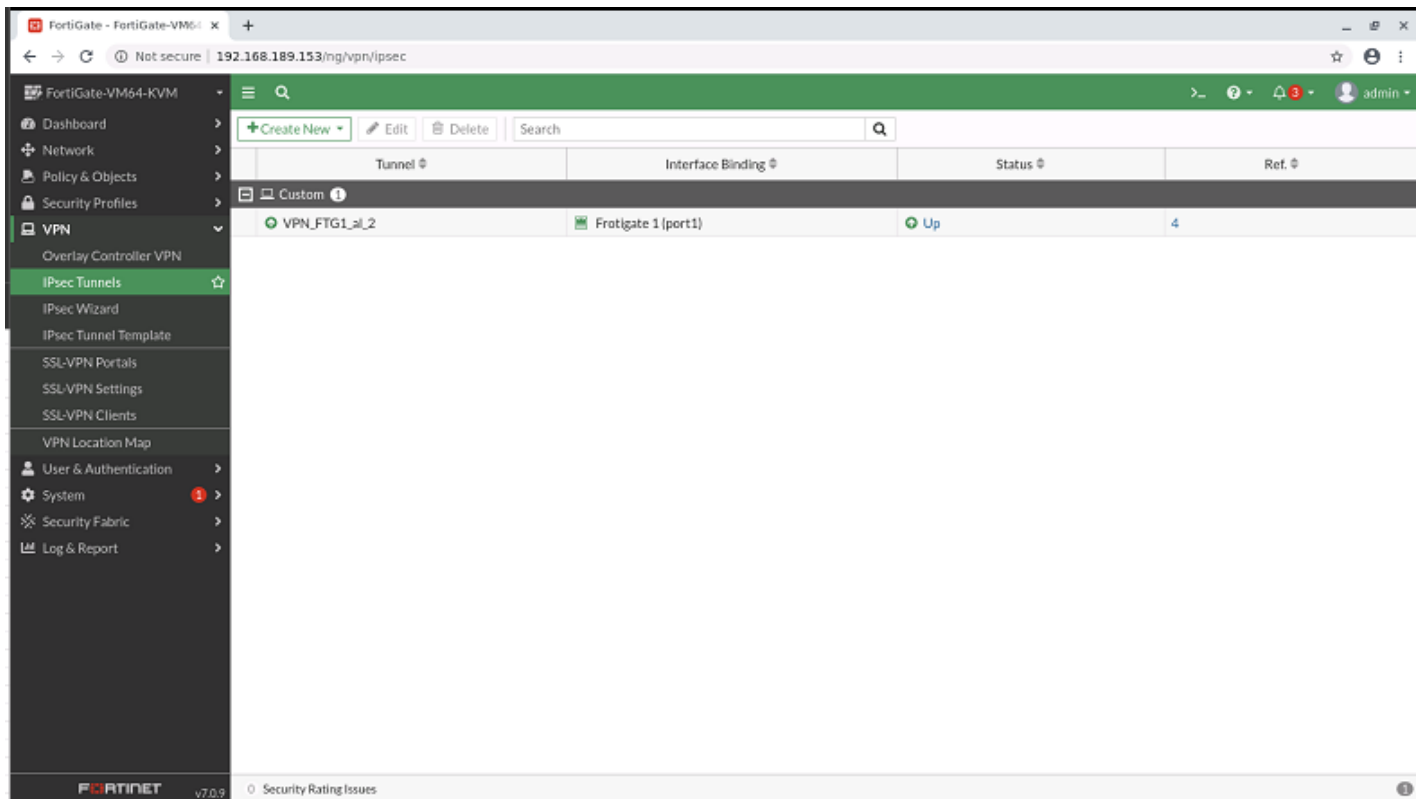
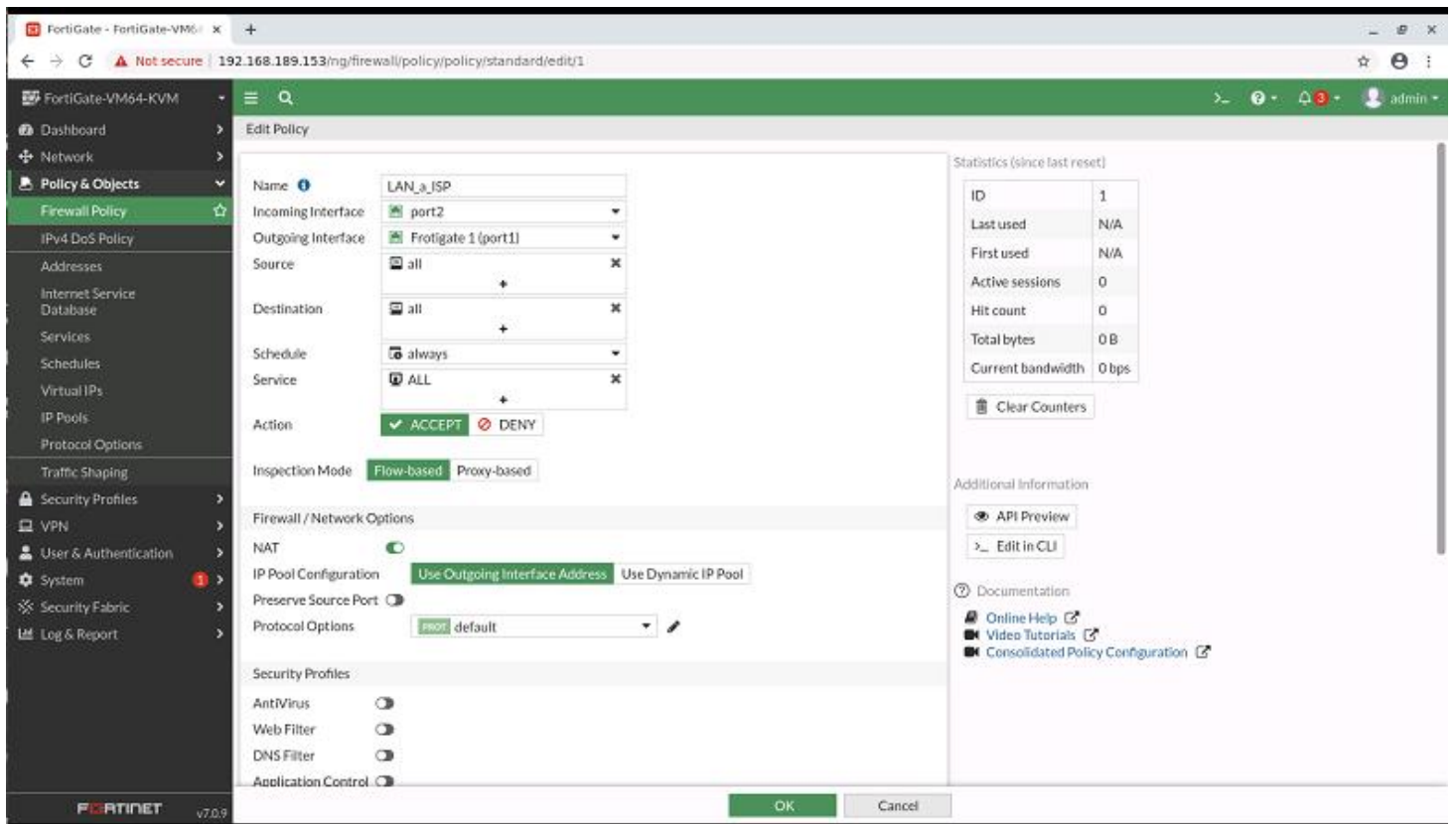
FortiGate-VM64-KVM

FortiGate VM64-KVM

Create New Edit Delete Integrate Interface Search Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2
Physical Interface							
Frotigate 1 (port1)	Physical Interface		14.10.1.2/255.255.255.252	PING HTTPS SSH HTTP FMG-Access			4
port2	Physical Interface		14.10.10.1/255.255.255.128	PING	1	14.10.10.2-14.10.10.126	4
port3	Physical Interface		0.0.0.0/0.0.0.0				0
port4	Physical Interface		192.168.189.153/255.255.255.0	PING HTTPS SSH HTTP			0
port5	Physical Interface		0.0.0.0/0.0.0.0				0
Tunnel Interface							
NAT interface (nafroot)	Tunnel Interface		0.0.0.0/0.0.0.0				0





FortiGate - FortiGate-VM64-KVM

192.168.189.153/mg/system/dashboard-preview/ipsec-vpn?locate=%7B%22name%22%3A%22VPN_FTG1_al_2%22%7D

admin

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

IPsec

Reset Statistics

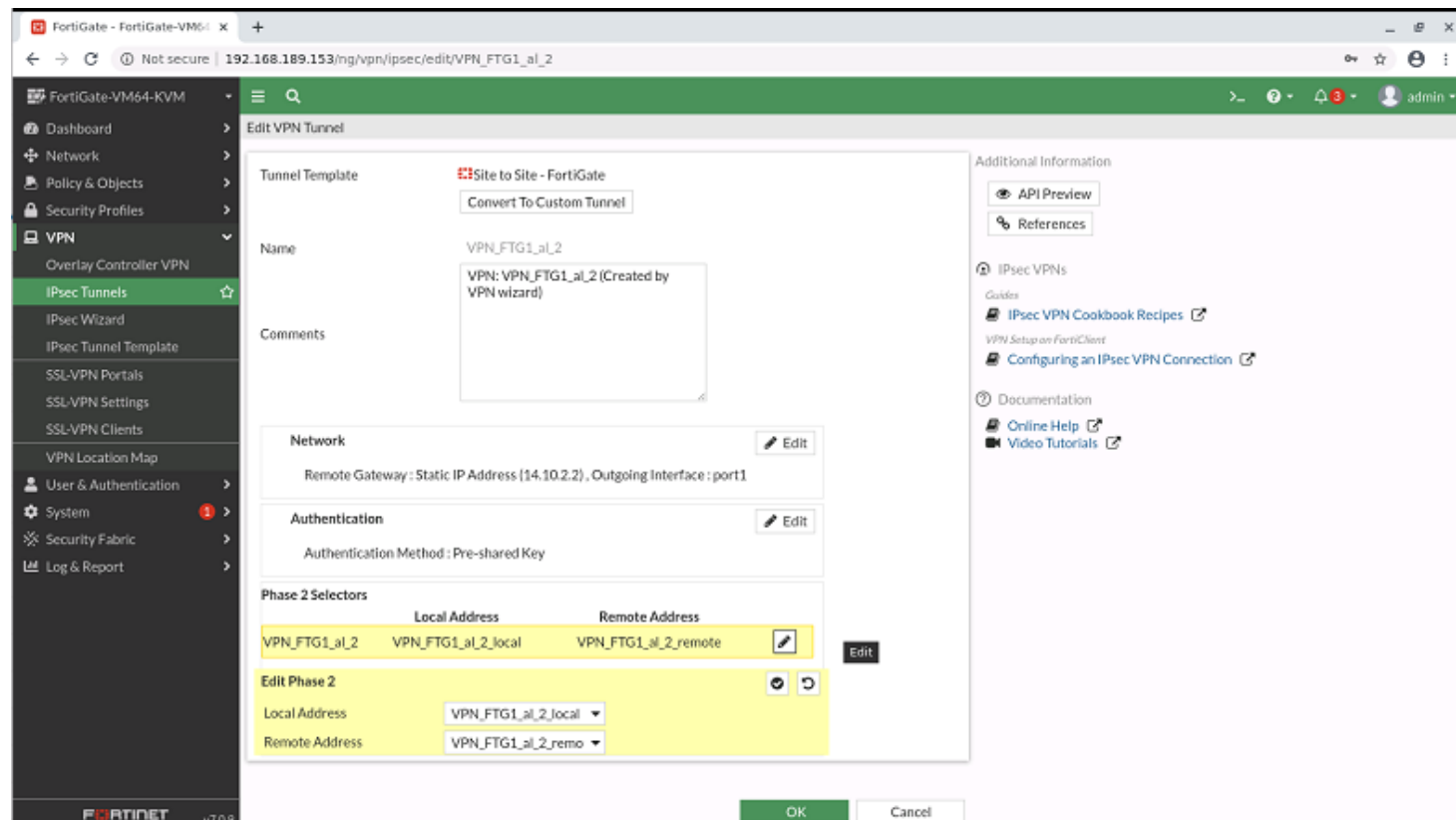
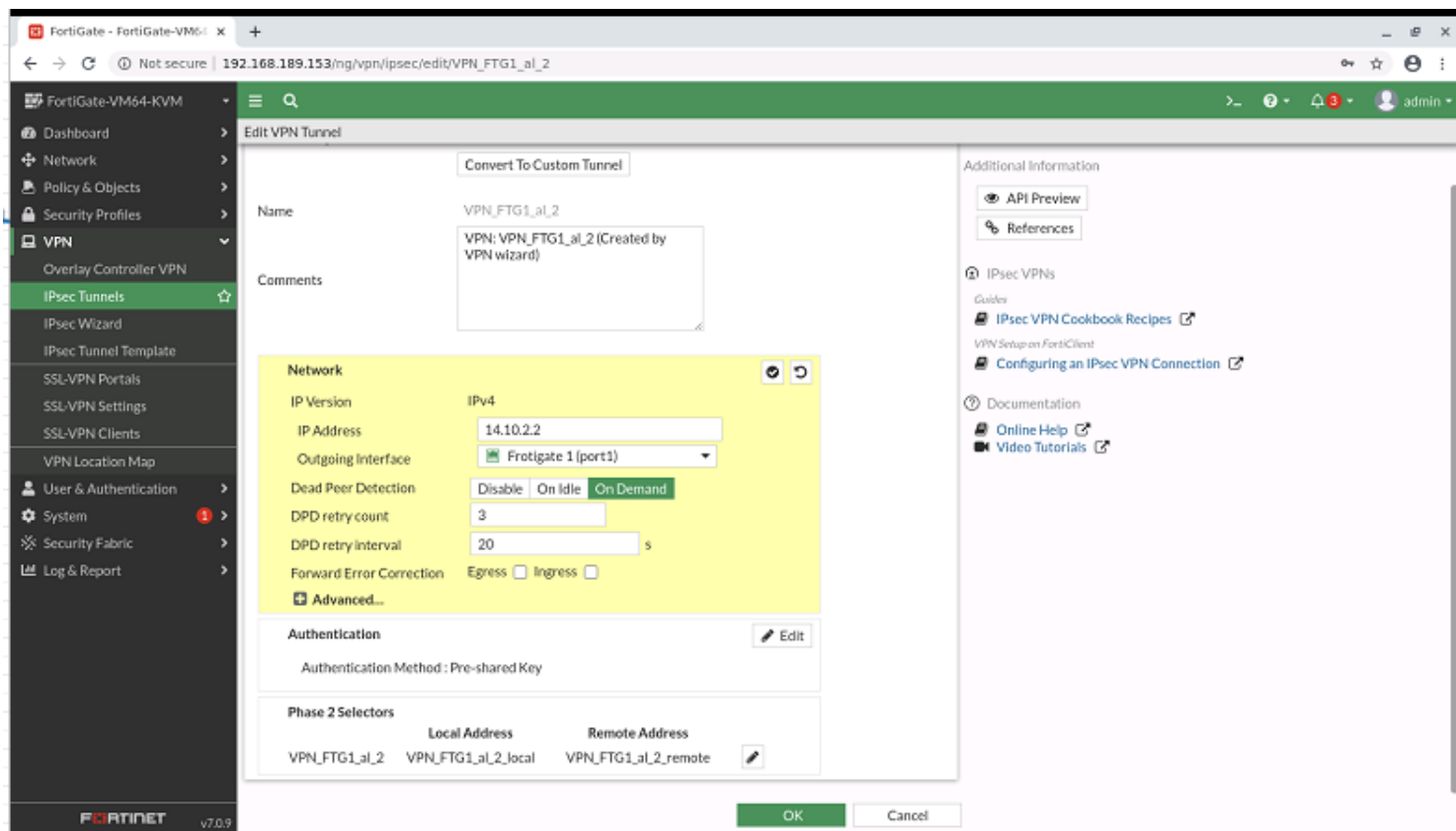
Bring Up

Bring Down

Locate on VPN Map

Name	Remote Gateway	Peer ID	Incoming Data	Outgoing Data	Phase 1	Phase 2 Selectors
VPN_FTG1_al_2	14.10.2.2		420 B	420 B	VPN_FTG1_al_2	VPN_FTG1_al_2

Updated: 19:24:49



Fortigate 2

FortiGate - FortiGate-VM64-KVM x +

Not secure 192.168.189.155/ng/interface

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23
2 4 6 8 10 12 14 16 18 20 22 24

+ Create New Edit Delete Integrate Interface Search Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2
Physical Interface							
fortigate 2 (port1)	Physical Interface		14.10.2.2/255.255.255.252	PING HTTPS SSH HTTP FMG Access			4
VPN_FTG2_a_1	Tunnel Interface		0.0.0.0/0.0.0.0				4
port3	Physical Interface		0.0.0.0/0.0.0.0				0
port4	Physical Interface		192.168.189.155/255.255.255.0	PING HTTPS SSH HTTP			0
Servers (port2)	Physical Interface		14.10.20.1/255.255.255.240	PING			3
Tunnel Interface							
NAT interface (natroot)	Tunnel Interface		0.0.0.0/0.0.0.0				0

Security Rating Issues Updated: 14:50:08

FortiGate - FortiGate-VM64-KVM x +

Not secure 192.168.189.155/ng/firewall/policy/policy/standard

FortiGate VM64-KVM

+ Create New Edit Delete Policy Lookup Search Export Interface Pair View By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
LAN_a_JSP									
Servers (port2) -- fortigate 2 (port1)	all	all	always	ALL	ACCEPT	Enabled	no-inspection	UTM	0B
Servers (port2) -- VPN_FTG2_a_1									
vpn_VPN_FTG2_a_1_local_0	VPN_FTG2_a_1_local	VPN_FTG2_a_1_remote	always	ALL	ACCEPT	Disabled	no-inspection	UTM	0B
VPN_FTG2_a_1 -- Servers (port2)									
vpn_VPN_FTG2_a_1_remote_0	VPN_FTG2_a_1_remote	VPN_FTG2_a_1_local	always	ALL	ACCEPT	Disabled	no-inspection	UTM	0B
Implicit									

Security Rating Issues Updated: 14:50:48

FortiGate - FortiGate-VM: x +

Not secure 192.168.189.155/ng/firewall/policy/policy/standard/edit/1

FortiGate-VM64-KVM

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate-VM64-KVM v7.0.9

Edit Policy

Name: LAN_a_JSP

Incoming Interface: Servers (port2)

Outgoing Interface: fortigate 2 (port1)

Source: all

Destination: all

Schedule: always

Service: ALL

Action: ☒ ACCEPT ☐ DENY

Inspection Mode: ☒ Flow-based ☐ Proxy-based

Firewall / Network Options

NAT: ☒ NAT

IP Pool Configuration: ☒ Use Outgoing Interface Address ☐ Use Dynamic IP Pool

Preserve Source Port: ☐ Preserve Source Port

Protocol Options: ☒ default

Security Profiles

AntiVirus: ☐ AntiVirus

Web Filter: ☐ Web Filter

DNS Filter: ☐ DNS Filter

Statistics (since last reset)

ID	1
Last used	N/A
First used	N/A
Active sessions	0
Hit count	0
Total bytes	0B
Current bandwidth	0bps

[Clear Counters](#)

Additional Information

[API Preview](#)

[Edit in CLI](#)

Documentation

[Online Help](#)

[Video Tutorials](#)

[Consolidated Policy Configuration](#)

OK Cancel

FortiGate - FortiGate-VM: x +

Not secure 192.168.189.155/ng/routing/static

FortiGate-VM64-KVM

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate-VM64-KVM v7.0.9

Static Routes

[+ Create New](#) [Edit](#) [Clone](#) [Delete](#)

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0	14.10.2.1	fortigate 2 (port1)	Enabled	
VPN_FTG2_a_1_remote	14.10.1.2	VPN_FTG2_a_1	Enabled	VPN: VPN_FTG2_a_1 (Created by VPN wizard)
VPN_FTG2_a_1_remote		Blackhole	Enabled	VPN: VPN_FTG2_a_1 (Created by VPN wizard)

FortiGate - FortiGate-VM: x

Not secure 192.168.189.155/ng/vpn/ipsec

FortiGate-VM64-KVM

- Dashboard
- Network
- Policy & Objects
- Security Profiles
- VPN
 - Overlay Controller VPN
 - IPsec Tunnels
 - IPsec Wizard
 - IPsec Tunnel Template
 - SSL-VPN Portals
 - SSL-VPN Settings
 - SSL-VPN Clients
 - VPN Location Map
- User & Authentication
- System
- Security Fabric
- Log & Report

IPsec Tunnels

+ Create New Edit Delete Search

Tunnel	Interface Binding	Status	Ref
Custom			
VPN_FTG2_a_1	fortigate 2 (port1)	Up	4

FortiGate v7.0.9

Security Rating Issues

FortiGate - FortiGate-VM: x

Not secure 192.168.189.155/system/dashboard-preview/ipsec-vpn?locate=%7B%22name%3A%22%22%3A%22VPN_FTG2_a_1%22%7D

FortiGate-VM64-KVM

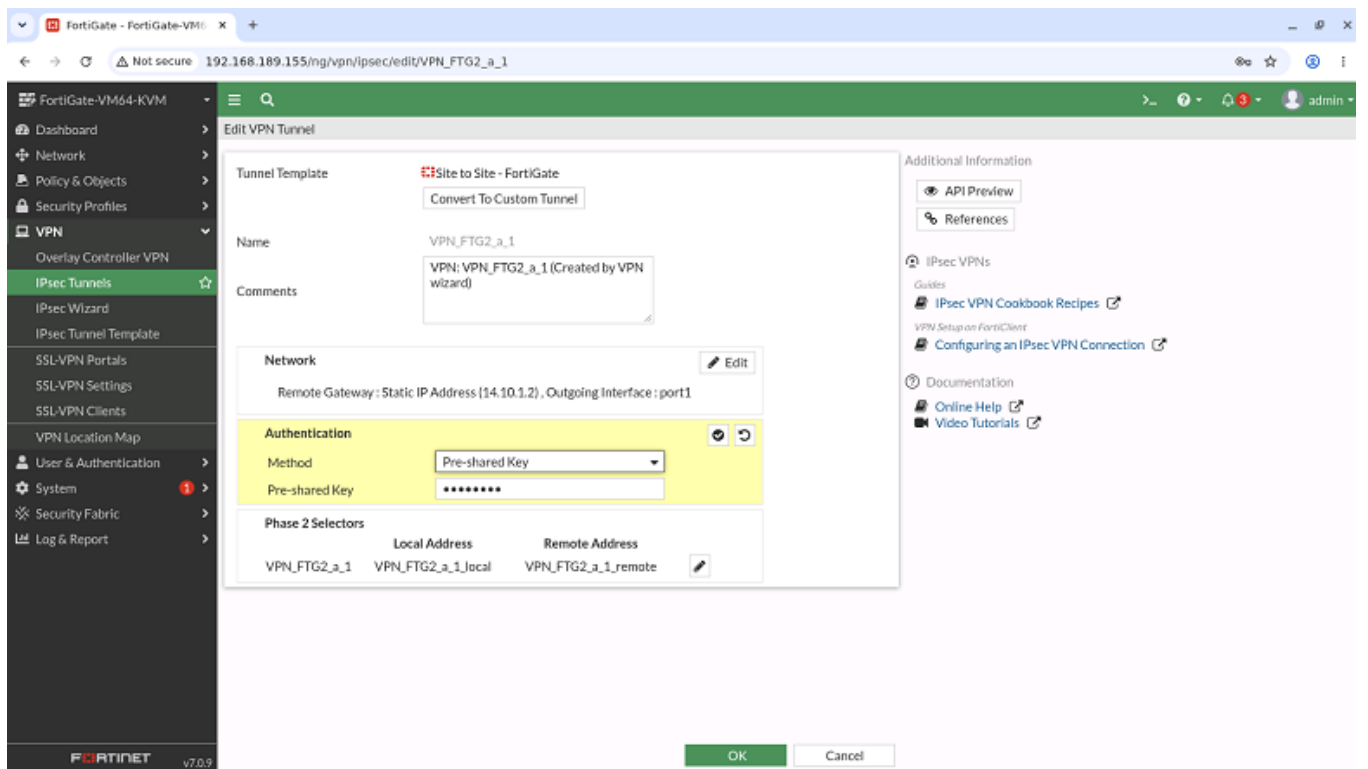
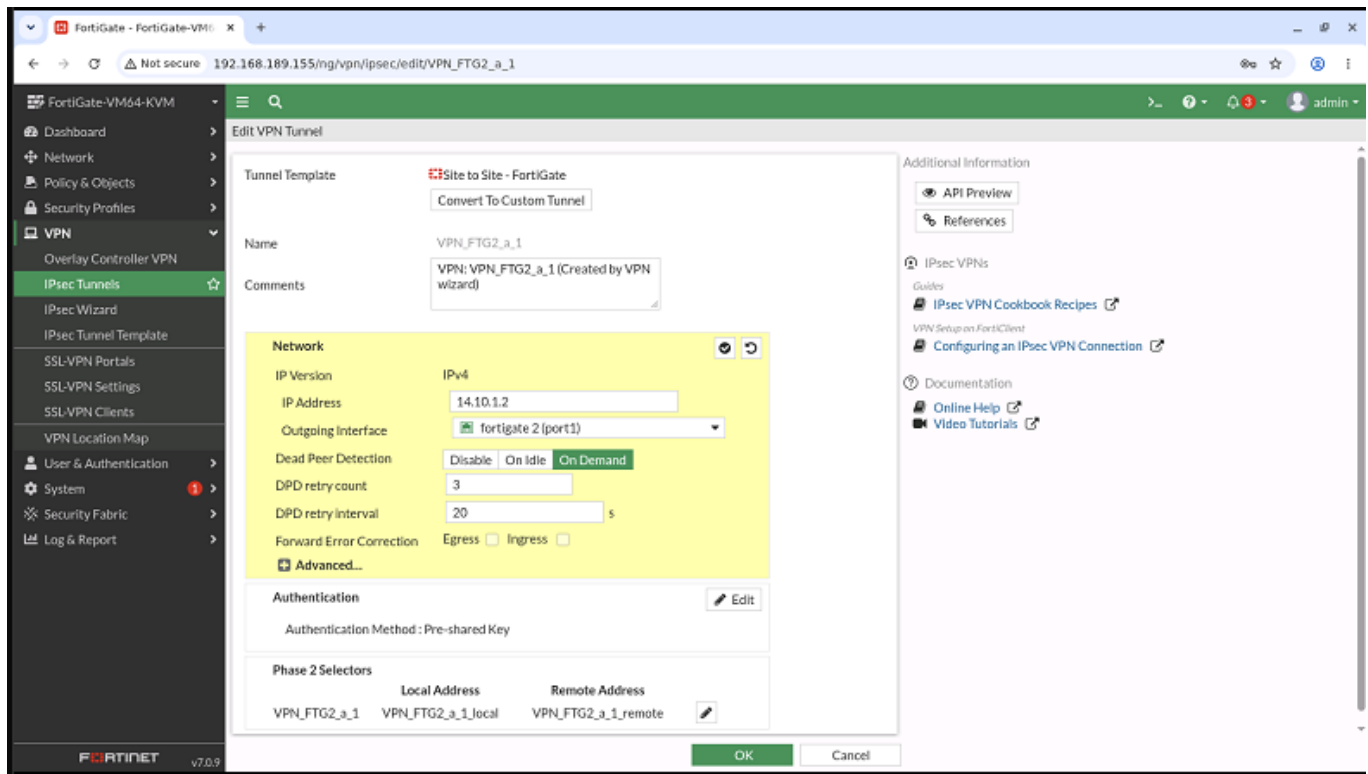
- Dashboard
- Network
- Policy & Objects
- Security Profiles
- VPN
 - Overlay Controller VPN
 - IPsec Tunnels
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 - SSL-VPN Portals
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 - SSL-VPN Clients
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- System
- Security Fabric
- Log & Report

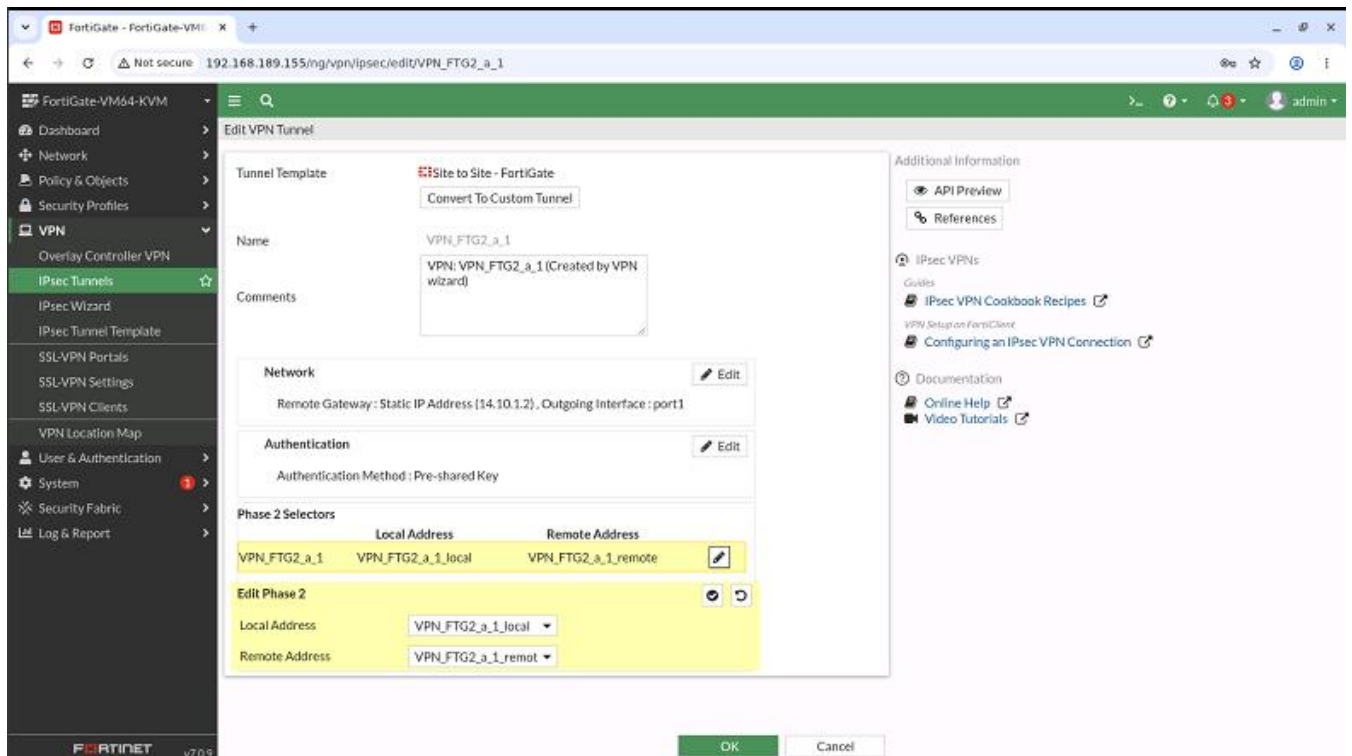
IPsec

Reset Statistics Bring Up Bring Down Locate on VPN Map

Name	Remote Gateway	Peer ID	Incoming Data	Outgoing Data	Phase 1	Phase 2 Selectors
Custom						
VPN_FTG2_a_1	14.10.1.2		420 B	420 B	VPN_FTG2_a_1	VPN_FTG2_a_1

Updated: 19:26:15





```

root@Docker:/home# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
318: eth0@if319: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 02:42:0a:b1:00:02 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.177.0.2/16 brd 10.177.255.255 scope global eth0
        valid_lft forever preferred_lft forever
321: docker17_0@if320: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 50:00:00:11:00:00 brd ff:ff:ff:ff:ff:ff link-netnsid 0
323: eth1@if322: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 50:00:00:11:00:01 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 14.10.20.2/28 scope global eth1
        valid_lft forever preferred_lft forever
325: eth2@if324: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state LOWERLAYERDOWN group default qlen 1000
    link/ether 50:00:00:11:00:02 brd ff:ff:ff:ff:ff:ff link-netnsid 0
root@Docker:/home#

```

```

root@Docker:/home#
root@Docker:/home# curl -k https://14.10.20.2

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2016-11-16
    See: https://launchpad.net/bugs/1288690
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

```

```
VPC
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DDORA IP 14.10.10.2/25 GW 14.10.10.1

VPCS> ping 14.10.20.2

14.10.20.2 icmp_seq=1 timeout
84 bytes from 14.10.20.2 icmp_seq=2 ttl=62 time=3.001 ms
84 bytes from 14.10.20.2 icmp_seq=3 ttl=62 time=1.555 ms
84 bytes from 14.10.20.2 icmp_seq=4 ttl=62 time=1.687 ms
84 bytes from 14.10.20.2 icmp_seq=5 ttl=62 time=2.392 ms

VPCS> 
```

```
VPCS> trace 14.10.20.2
trace to 14.10.20.2, 8 hops max, press Ctrl+C to stop
 1  14.10.10.1    0.339 ms  0.270 ms  0.234 ms
 2  14.10.2.2    0.938 ms  0.594 ms  0.717 ms
 3  *14.10.20.2  1.203 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> 
```